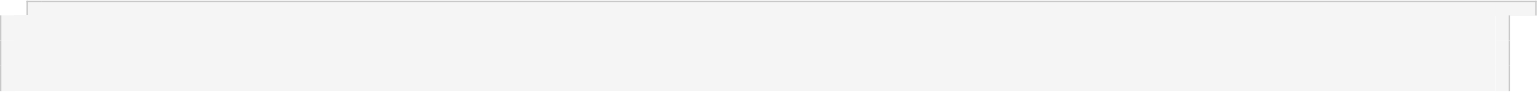
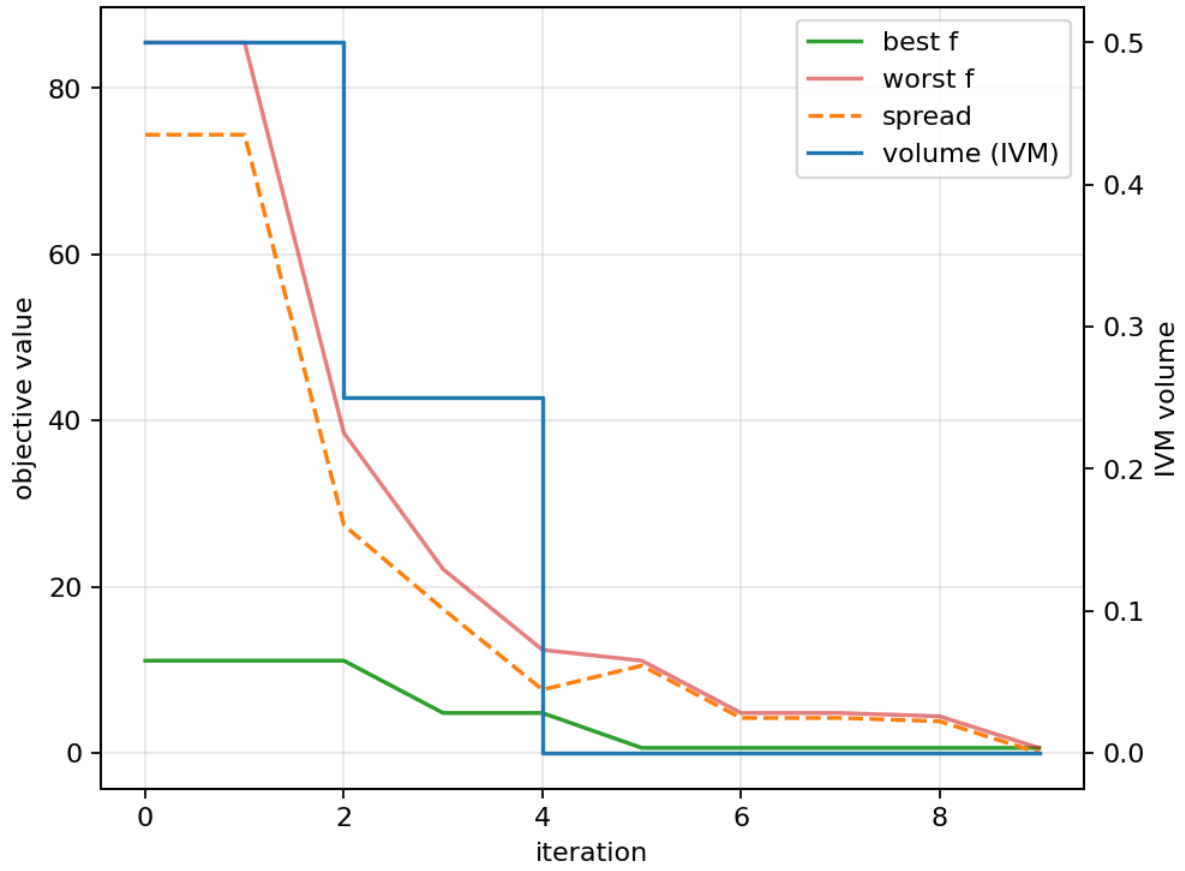


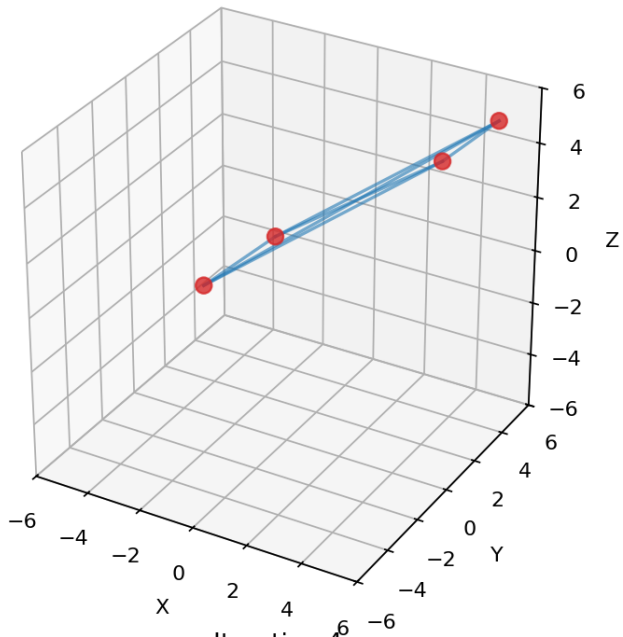
$\alpha \approx 1$
 $\gamma \approx 2$
 $\rho \approx 0.5$
 $\sigma \approx 0.5$



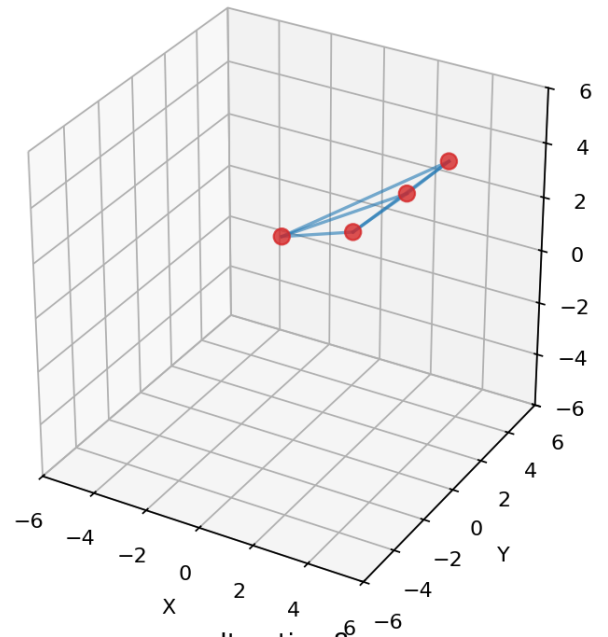


Nelder-Mead Simplex Evolution on Integer Quadray Lattice

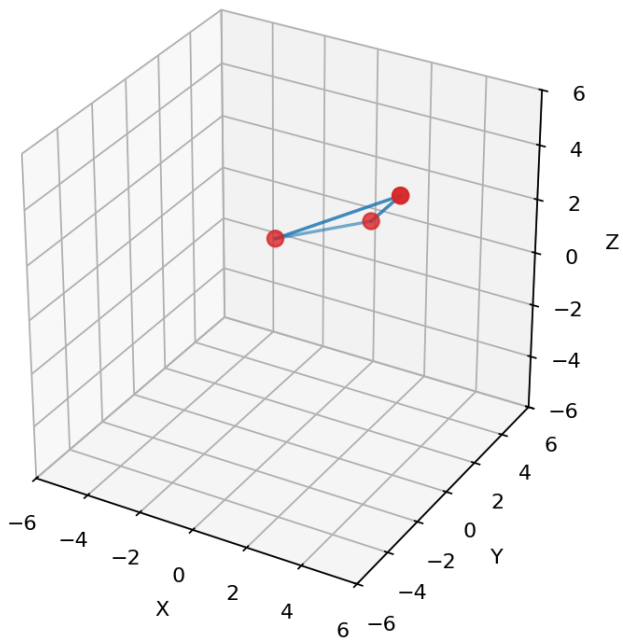
Iteration 0
Best: 11.1, Spread: 74.4



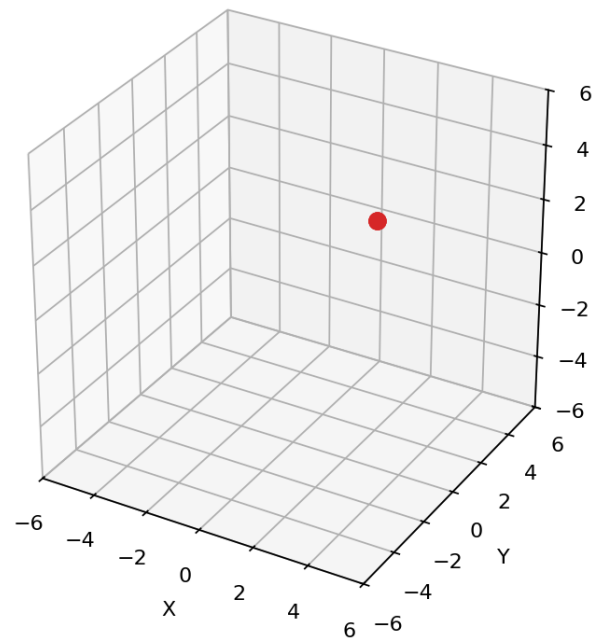
Iteration 2
Best: 4.8, Spread: 17.3



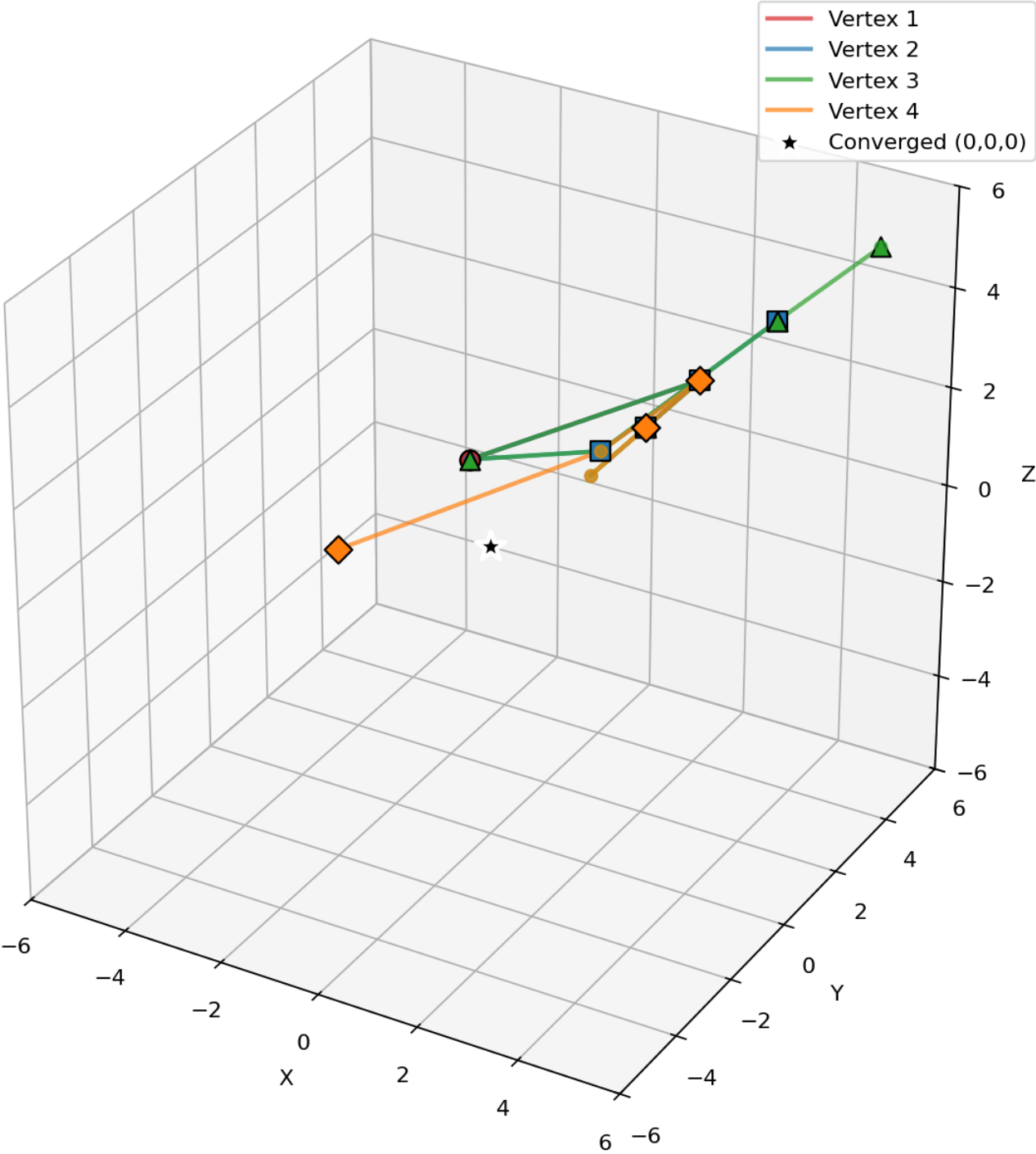
Iteration 4
Best: 0.6, Spread: 10.5



Iteration 8
Best: 0.6, Spread: 0.0



Complete Simplex Optimization Trace Nelder-Mead on Integer Quadray Lattice



$$F_{ij}$$

$$=(w_0,w_1,w_2)L()$$

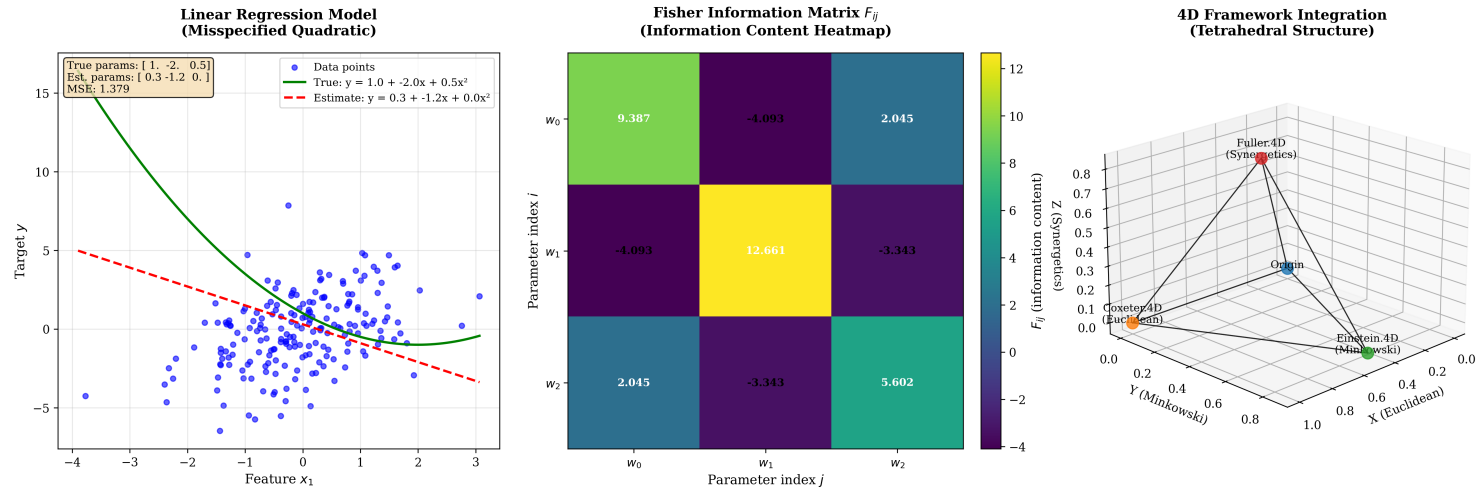
$$L_n$$

$$g_{ij}=F_{ij}\delta_{ij}F^{-1}\nabla L\nabla L$$

$$F^{-1}\nabla L$$

$$\delta_{ij}F_{ij}$$

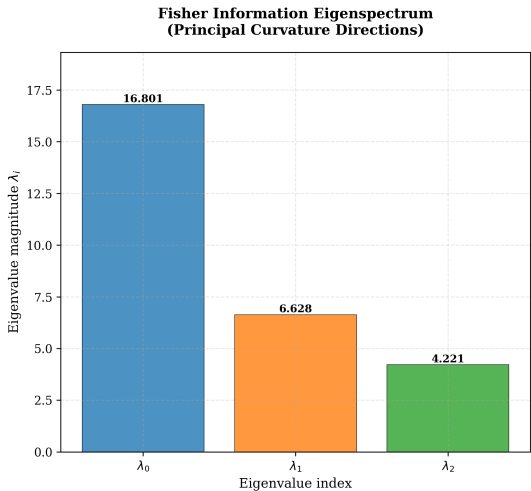
$$F^{-1}\nabla L\Delta=-\eta F^{-1}\nabla L$$



$$y=w_0+w_1x+w_2x^2w_{true}=[1.0,-2.0,0.5]w_{est}=[0.3,-1.2,0.0]F_{ij}F_{ii}w_iF_{ij}w_0,w_1,w_2\delta_{ij}F_{ij}\Delta w=F^{-1}\nabla L$$

$$\eta F\nabla Lg_{ij}=F_{ij}$$

$$F^{-1}\nabla L$$



Curvature Analysis & 4D Framework

Key Metrics:	
Condition Number:	3.98
Anisotropy Index:	0.592
Total Curvature:	27.651

Eigenvalue Interpretation:

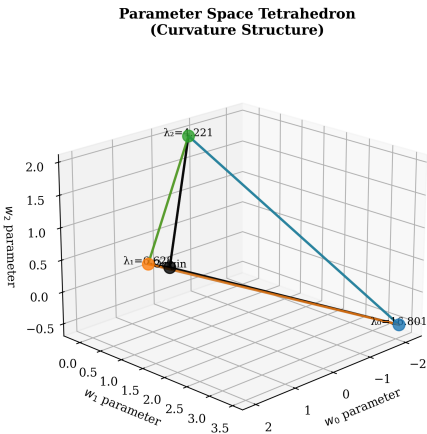
- $\lambda_0 = \therefore 3\%$ Primary direction
- $\lambda_1 = \therefore 3\%$ Secondary direction
- $\lambda_2 = \therefore 3\%$ Tertiary direction

4D Framework Connection:

- Coxeter.4D: Euclidean parameter space
- Einstein.4D: Fisher metric geometry
- Fuller.4D: Tetrahedral structure

Optimization Implications:

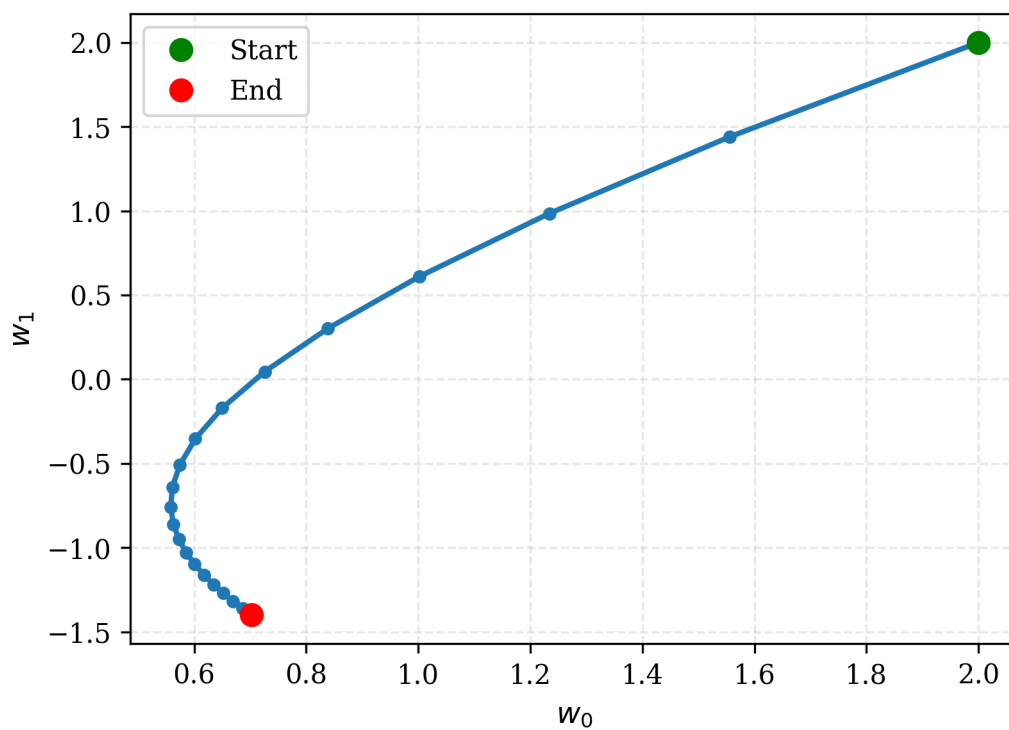
- Natural gradient scales by F^{-1}
- Anisotropic scaling improves convergence



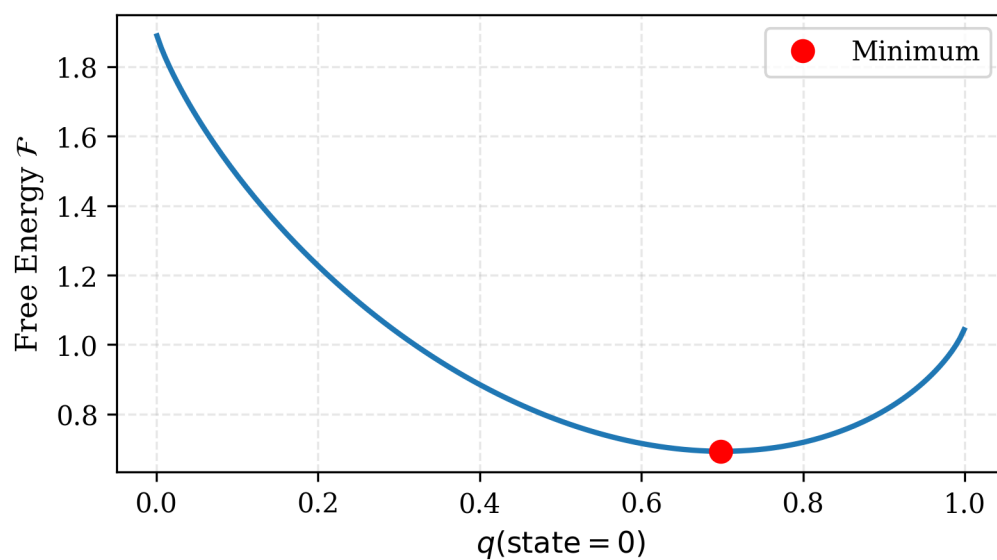
$\lambda_i F^{-1} F^{-1}$

F

Natural Gradient Trajectory (Geodesic Motion on Information Manifold)



Variational Free Energy Landscape (2-State System)



$$\mathcal{F} = -P(o|s) + [Q(s)||P(s)]q(=0)\mathcal{F}$$